

AI-based Village Pond Planning System

CSD Assignment 1 – Final Technical Report

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Village ponds harvest monsoon runoff, but siting one needs terrain, drainage, rainfall and storage analysis that a Gram Panchayat rarely has. We built a web application in which an administrator either draws a box around the village land on a satellite map or uploads a KML/KMZ contour map, and receives a ranked pond location, the catchment draining to it, the expected water volume and a costed pond design, all overlaid on the map. Elevation comes from the upload (Delaunay TIN) or from the Copernicus GLO-30 DEM for the drawn box; both feed one validated hydrology chain: Priority-Flood conditioning, D8 flow routing as a forest of in-trees, a single topological sweep for accumulation and for a complete-catchment test, multi-criteria site scoring with hard exclusions for rivers and mapped water, and catchment delineation by reverse breadth-first search. Runoff is SCS-CN on 45 years of daily ERA5-Land rainfall; the pond is sized by cost and checked by a daily water balance. Every number carries a unit and an uncertainty band. The catchment engine matches the independent pysheds implementation within 2–3 % on the same DEM; 219 tests (41 golden) pass; 46/46 end-to-end HTTP checks pass on the deployed lab VM. On one process a village-sized selection is analysed in 2 s (median) under 20 concurrent users.

Additional Key Words and Phrases: Village Pond Planning, Geospatial Analysis, Rainwater Harvesting, Catchment Estimation, D8 Flow Routing, Web GIS, REST APIs

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1 INTRODUCTION

Rural India stores monsoon water in small excavated ponds and tanks. A pond is only useful where the terrain delivers enough runoff to fill it and not so much that the embankment becomes a river dam. This report describes a system that finds such places from public geospatial data and explains its answer. Section 2 maps the requirements to modules; Sections 3–4 give the high- and low-level design and the algorithms (the examined core); Sections 5–6 cover implementation and the CSD themes; Section 7 evaluates; Section 8 states the limits.

1.1 Motivation

Selecting a site by hand needs a topographic survey, a rainfall record, a runoff estimate and a storage calculation. Contour sheets are not digitised for most villages, rainfall gauges are sparse, and land records are not public. The questions, however, are computable: where does water converge, how much land drains there, how much rain falls in a normal (75 % dependable) year, and what pond does that fill? An automated tool that answers them in seconds, with stated uncertainty, lets an administrator shortlist sites before paying for a survey.

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1.2 Scope of the Project

In scope: selecting a land area on a map (a drawn box) or uploading a KML/KMZ contour map; terrain products (DEM, slope, TWI, contours, streams); ranked pond sites with per-criterion scores; catchment of any clicked point; 45-year rainfall statistics; runoff by three methods; pond depth, dimensions, storage, fill reliability, spillway and indicative cost; land-availability and water masks; saving, approving and exporting a recommendation. **Out of scope:** structural design of the embankment, groundwater recharge modelling, legal land ownership (never assumed; shown as unknown), and survey-grade accuracy — all results are planning grade from 30 m public DEMs.

2 PROBLEM STATEMENT AND REQUIREMENTS

The assignment asks for a web application that analyses terrain and rainfall to recommend pond sites and estimates catchment, runoff, depth and storage on an interactive map. The Phase 3 brief adds: select the land area on a map; return the suggested pond location, catchment area and expected water volume for it, overlaid on the map; stay fast within the four provided lab systems. Table 1 maps each requirement to its module.

Table 1. Functional requirements and where they are implemented

Requirement	Implemented in
Select land area on a map (Phase 3)	web/src/components/AreaPanel.tsx, MapView.tsx; POST /analyzeArea; engines/terrain/adapters.py (ProviderTileAdapter)
Satellite imagery display	Esri World Imagery basemap in MapView.tsx; village named by reverse geocoding (providers/geocoding.py)
Contour map visualization	engines/terrain/contours.py (marching squares + Douglas-Peucker); GET /terrain/{id}/contours
Available-land identification	engines/suitability/constraints.py (Specification pattern), water_mask.py (NDWI + OpenCV)
Catchment area estimation	engines/hydrology/{conditioning, flow, catchment}.py; POST /analysis/catchment
Suggested pond location	engines/hydrology/siting.py (MCDA + hard exclusions + NMS)
Historical rainfall query	providers/rainfall/ (Open-Meteo ERA5-Land, NASA POWER), engines/rainfall/statistics.py
Runoff volume estimation	engines/runoff/ (SCS-CN on the daily series, rational, Strange)
Pond depth / storage recommendation	engines/design/ (PondDesignBuilder, EAV, water balance, spillway)
Combined overlay / results view	web/src/components/ResultsOverlay.tsx + map layers

2.1 Non-Functional Requirements

Performance: a village-sized area (0.25–25 km²) analysed end to end in under 10 s; a catchment click under 2 s. **Scalability and stress:** long work runs as asynchronous jobs on bounded pools (bulkheads); a full queue answers 429 rather than queueing without bound; request size is capped by area; replicas scale horizontally behind a load balancer. **Availability:** external providers are wrapped in retry, circuit breaker, cache and fallback; a dead replica is removed from rotation; the frontend works offline from cache. **Correctness and honesty:** every number carries a unit and an uncertainty band; warnings name every assumption. **Security:** JWT (RS256) with viewer/planner/officer roles on the approval workflow; append-only audit log. **Usability:** works on current Chrome/Firefox/Safari and at 390 px width; Lighthouse accessibility 100.

3 SYSTEM ARCHITECTURE AND HIGH-LEVEL DESIGN

The system is a layered monolith (ADR 0001) behind asynchronous job routes (Figure 1). The browser talks to one origin. The API validates and enqueues; a worker runs the use-case workflow, which sequences pure engines and calls external data only through adapters. Dependencies point inward only — api → engines → providers/repositories → domain — and a test (tests/test_layering.py) fails the build if a router holds business logic or an engine imports a framework.

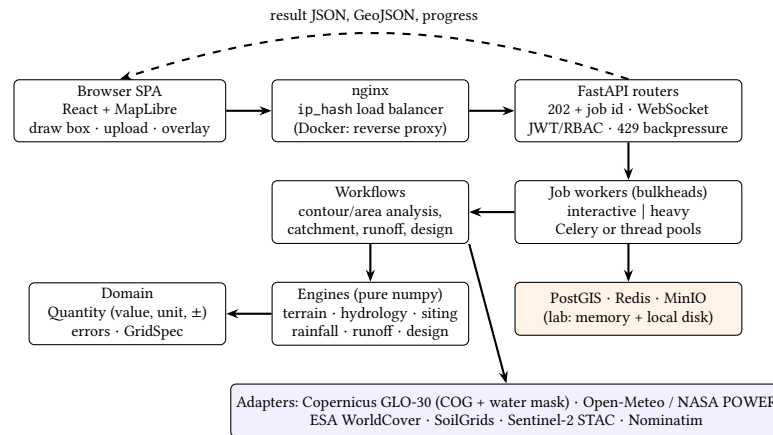


Fig. 1. High-level system architecture. Arrows show the request path of an analysis: the UI posts a box or a file, receives 202 with a job id, and follows progress over a WebSocket (polling fallback) until the result is overlaid on the map.

The same code runs in two shapes. The reference deployment is an 11-service Docker Compose stack (PostGIS, Redis, MinIO, API, two Celery workers, beat, TiTiler, nginx + SPA, Prometheus, Grafana). The four lab machines are unprivileged containers (no Docker, no systemd, no root), so there each replica is one process with in-memory persistence, a thread-pool job runner and a local object store, and replicas sit behind a user-space nginx load balancer. The switch is configuration only: persistence, job runner and object store are ports with two or three adapters each (ADR 0013).

3.1 Technology Stack

Python 3.12, FastAPI, Pydantic v2; numpy/scipy for every algorithm (no numba: at village scale a legible sorted loop is fast enough); rasterio/GDAL for COG window reads and reprojection; pyproj for UTM; OpenCV for the NDWI water mask; PostgreSQL + PostGIS, Redis + Celery, MinIO; React + TypeScript + MapLibre GL; Docker Compose; GitHub Actions. Elevation: the uploaded contour map, or Copernicus GLO-30 COGs [5] for a drawn box. Rainfall: Open-Meteo ERA5-Land [9] with NASA POWER as fallback. **Deviations from the suggested stack, justified:** FastAPI over Flask (typed OpenAPI contract generated from the code, dependency injection for ports); PostgreSQL + PostGIS over MongoDB (spatial types, boundary reuse by ST_Equals, append-only audit rules in the database); full DEM rasters over point-elevation APIs (flow direction needs a grid — a village is ~10 000 cells, one COG window read versus thousands of throttled point calls); ERA5-Land over IMD (keyless, daily, 45 years in 2 s; IMD gridded data needs registration and manual download).

3.2 Low-Level Design

- **Ports and adapters.** DEMProvider has two adapters: ContourKMLAdapter (parse, infer provenance, TIN) and ProviderTileAdapter (GLO-30 window read and warp). The analysis workflow's *only* branch on input kind is which adapter to build. JobRunner has three (Celery, thread pools, inline for tests). Each rainfall provider is wrapped as Cached(CircuitBreaker(Retry(p))) inside a FallbackChain.
- **Contract.** Every analysis route returns JobAccepted{job_id, poll_url}. Result schemas (ContourAnalysisResult, CatchmentResult, PondDesignResult) are Pydantic models from which openapi.json and the frontend's TypeScript types are generated. Every numeric field is a Quantity {value, unit, uncertainty_pct, low, high, method} — a domain type, so a bare float cannot reach the API.
- **Persistence of an analysis** is a Saga of four idempotent steps (village → rasters → streams → DEM asset) with compensations run in reverse on failure.
- **Data model:** villages (PostGIS boundary), dem_assets (provenance, statistics, object keys), jobs (status, stage, progress, result JSON, idempotency key), recommendations + outbox + append-only audit_log; rasters are Cloud-Optimised GeoTIFFs in the object store.

4 METHODOLOGY

Figure 4 (appendix) shows the pipeline; this section gives each stage.

4.1 Terrain and Elevation Analysis

Map-selected area. The box's centroid fixes the UTM zone; the grid is the envelope of the projected box at GLO-30's native 30 m. For every 1° tile the grid touches, the window under the grid's own lon/lat footprint (plus two source cells) is read from the COG over HTTP — a few hundred kB — and warped bilinearly onto the grid; tiles are mosaicked. The tile's Water Body Mask (lake/river classes) is warped by nearest neighbour onto the same grid. Limits of 0.25–25 km², measured in metres, return 422 area_out_of_range outside them. **Uploaded contour map.** The KML/KMZ is parsed namespace-agnostically; elevation is read by an ordered strategy (Z → whitelisted ExtendedData field → placemark name), rejecting numeric ID decoys — the provided sample has one. Vertices are densified and triangulated (Delaunay TIN, linear interpolation, Gaussian $\sigma=1$ cell). The cell size is derived, not configured: mean contour spacing (hull area / total contour length) / 4, floored at the source DEM's resolution, which is inferred from the file's own metadata (the sample: SRTM, 30 m). **Surfaces.** Slope, aspect and hillshade by Horn's 3×3 kernel [7]; profile and plan curvature [19]; topographic wetness index $\ln(a/\tan \beta)$ [3]; contours by marching squares, simplified by Douglas-Peucker [4]. **OpenCV** builds the existing-water mask from Sentinel-2 NDWI [8]: Otsu threshold [11], morphological open/close, connected components with an area filter.

4.2 Catchment Area Delineation

Conditioning. Priority-Flood with $\varepsilon [1, 17]$ fills every depression to its spill level and gives flats a tiny gradient in one $O(n \log n)$ pass, so every cell drains. **The flow graph.** D8 [10] gives each cell one out-edge to its steepest-descending neighbour (diagonal drops divided by $\sqrt{2}$). Every node has out-degree ≤ 1 and the filled surface has no cycles, so the graph is a forest of in-trees rooted at map-edge outlets, and *descending filled elevation is a topological order*. One sorted sweep therefore computes flow accumulation (subtree size) and, with OR in place of +, whether a cell's catchment reaches the map edge (Algorithm 1). Streams are cells draining ≥ 5 ha (an area, so it means the same at any resolution),

ordered by Strahler [14]. **Delineation.** A click is snapped to the *nearest* cell within 150 m that drains ≥ 2 ha — nearest, not largest, so a tributary click is not dragged onto the river — and the distance moved is shown. The catchment is the reverse breadth-first search from the outlet over the inverted D8 edges (Algorithm 2); area, longest flow path (deepest BFS level) and relief come from the same walk. Area uncertainty is max(15%, one cell ring around the perimeter). *Assumptions and limits:* D8 cannot split flow (D- ∞ is a documented stub); on very flat land a 2 m DEM error can redirect a small flow path (Section 7); the result is bounded by the selected area — hence the complete-catchment constraint below.

Algorithm 1 One topological sweep: accumulation and edge-fed flags — $O(n \log n)$

Require: filled DEM z , receiver $r(c)$ from D8 (-1 at outlets)

- 1: $acc(c) \leftarrow 1$; $edge(c) \leftarrow [c \text{ on the grid border or beside nodata}]$ **for all** c
- 2: **for** c **in** cells sorted by z descending **do** ▷ a topological order of the D8 forest
- 3: **if** $r(c) \neq -1$ **then**
- 4: $acc(r(c)) \leftarrow acc(r(c)) + acc(c)$ ▷ subtree size = upstream cells
- 5: $edge(r(c)) \leftarrow edge(r(c)) \vee edge(c)$ ▷ catchment reaches the map edge
- 6: **return** $acc \times$ cell area, $edge$

Algorithm 2 Catchment of a pour point — reverse BFS, $O(\text{catchment size})$

Require: outlet o after snapping; donor lists $D(c) = \{d : r(d) = c\}$ (inverse D8, built once)

- 1: $Q \leftarrow [o]$; $seen \leftarrow \{o\}$; $level(o) \leftarrow 0$
- 2: **while** Q not empty **do**
- 3: $c \leftarrow Q.POP()$
- 4: **for** $d \in D(c)$ **do** $seen \leftarrow seen \cup \{d\}$; $level(d) \leftarrow level(c) + \ell(d, c)$; $Q.PUSH(d)$
- 5: **return** mask $seen$, area $|seen| \cdot \text{cell area}$, longest flow path $\max level$

4.3 Site Selection — How the Area Is Chosen

The suggested location is the best-scoring cell of the drainage network after *hard exclusions* (Algorithm 3). Scoring is a weighted sum of four memberships in $[0, 1]$ with weights from a Saaty pairwise matrix (principal eigenvector, consistency ratio 0.004 [13]): *upstream area* as a plateau in $\log_{10}$ ha (0 at 1 ha, 1 from 10 to 150 ha, 0 at 1 000 ha — enough to fill a pond, not a river), *flatness* (1 on 0–3 % slope, 0 at 15 %), *wetness* (normalised TWI) and *impoundment efficiency* (volume held behind a nominal 2 m rise by an 8-connected flood fill on the upstream side, divided by its footprint — natural basins score high). Exclusions come first because a score cannot express “never”: channels beyond 1 000 ha and mapped water are rivers; nothing within a 200 m flood belt of a channel beyond 150 ha or of mapped water; and nothing whose catchment reaches the map edge, because its area and runoff would be lower bounds. Non-maximum suppression at 200 m keeps the top five distinct.

Algorithm 3 Pond site selection (terrain MCDA with hard exclusions)

Require: streams S , upstream area A , slope s , TWI, water mask W , edge-fed flags E

- 1: $river \leftarrow (S \wedge A \geq 1000 \text{ ha}) \vee W$; $major \leftarrow (S \wedge A \geq 150 \text{ ha}) \vee W$
- 2: $C \leftarrow S \wedge (s \leq 15\%) \wedge \neg river \wedge \text{DIST}(major) \geq 200 \text{ m}$ ▷ distance transform
- 3: **if** $C \wedge \neg E \neq \emptyset$ **then** $C \leftarrow C \wedge \neg E$ **else** warn no_complete_catchment
- 4: **for** $c \in C$ **do** $score(c) \leftarrow \sum_k w_k \mu_k(c)$ ▷ area plateau, flatness, TWI, impoundment
- 5: **return** top 5 by $score$ with non-maximum suppression (200 m), each with its μ_k

4.4 Rainfall Data Integration

Daily precipitation 1981–2025 at the site from Open-Meteo’s ERA5-Land archive, NASA POWER as fallback, both behind retry, circuit breaker, a single-flight object-store cache and a fallback chain that records which provider answered. Statistics use complete calendar years only: mean, CV, **75 % dependable rainfall by the Weibull plotting position** $P = m/(n+1)$ (the design year of Indian minor-irrigation practice), June–September share, IMD rainy days (≥ 2.5 mm), and the 25-year 1-day depth by Gumbel EV1 [6].

4.5 Runoff and Pond Sizing

Land cover from ESA WorldCover [18] and soil texture from SoilGrids [12] give an area-weighted TR-55 curve number [16]. SCS-CN $Q = (P - 0.2S)^2 / (P + 0.8S)$, $S = 25400/CN - 254$ mm, is applied to *each day* and summed — applied to annual totals it overestimates more than threefold (asserted by a test). The rational coefficient and Strange’s table [15] bracket it; the spread is reported. **Expected water volume** = 75 % dependable annual SCS-CN runoff $\times$ catchment area. Design: target = runoff $\times$ 0.6 harvest efficiency, clamped to 2 000–50 000 m³; an excavated frustum (2:1 sides, 0.5 m free-board) whose depth (1.5–3.5 m) and aspect are chosen by minimum cost; a daily water balance (inflow, evaporation, 2 mm/day seepage) gives the share of years the pond fills; the spillway passes the 25-year storm.

5 IMPLEMENTATION

5.1 Backend and API Design

REST under /api/v1, OpenAPI 3.1 generated from the code (Swagger at /docs). Key routes:

POST /analyzeArea	box $\rightarrow$ 202 job: site, catchment, terrain from GLO-30
POST /analyzeContour	KML/KMZ (contour_map) $\rightarrow$ 202 job, same result shape
GET /jobs/{id} · WS /jobs/{id}/ws	status, stage, percent
GET /analysis/results/contour/{job}	suggested location, rationale, catchment, 5 ranked sites, method
POST /analysis/catchment	catchment of any point (snap, BFS, metrics)
GET /rainfall/statistics	45-year statistics at a point
POST /analysis/runoff · /pond-design	runoff range; full design with reliability and cost
POST /analysis/suitability	land constraints, NDWI water mask, AHP ranking
GET /terrain/{id}/{contours,streams,layers}	map layers
/recommendations, /auth/token	draft $\rightarrow$ submitted $\rightarrow$ approved (RBAC), audit, PDF/GeoJSON/CSV

Errors are RFC 9457 problem documents with stable codes generated from the same table the handlers use (GET /meta/errors). Third-party failures degrade, never crash: rainfall falls back provider $\rightarrow$ cache $\rightarrow$ recorded record; land cover and soil fall back to stated defaults with a warning and a lower confidence label; an unreachable water mask or geocoder is skipped and said so. Authentication: RS256 JWT, roles viewer, planner, officer; approvals require officer.

5.2 Frontend and Visualization

A React + MapLibre single-page app (Figure 2). *Select land area*: search a place, click two opposite corners; the box’s area is checked live against the limits. *Analyse* posts the box, shows the job’s stage and percentage, then draws the selected area, contours with labels, streams by Strahler order, the five ranked sites, the top site’s catchment polygon and outlet, and — after the pond design runs automatically on the top site — the pond footprint and a results overlay with

location, catchment, 75 % rainfall, SCS-CN runoff, storage, dimensions, fill reliability and a confidence badge. Any map click delineates that point's catchment. Nineteen layers are toggleable; an offline-first service worker serves cached results under an "Offline" badge.

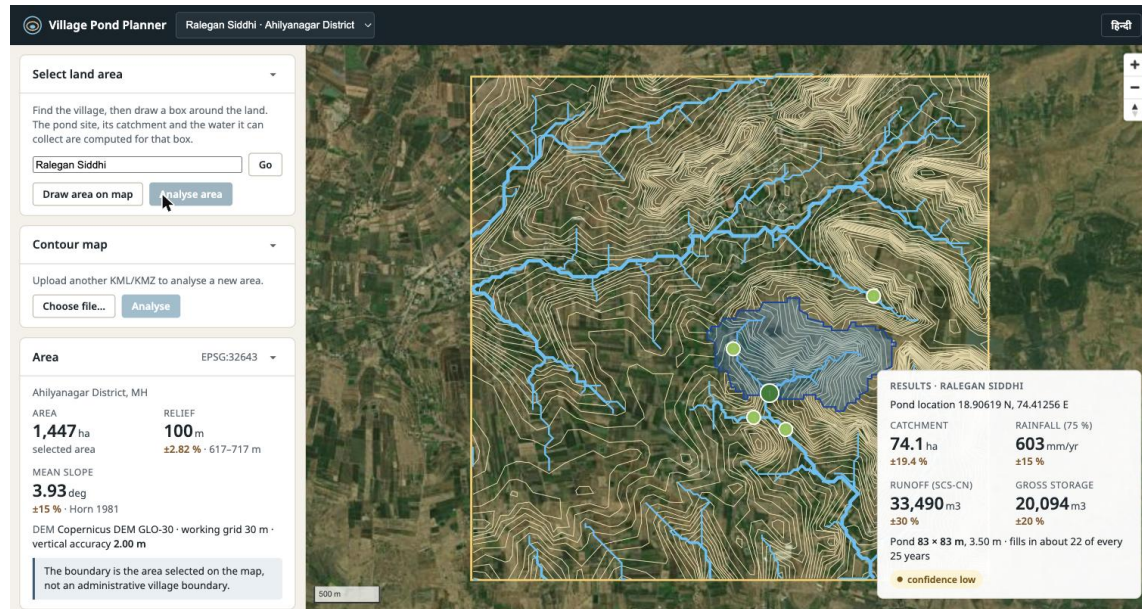


Fig. 2. Application interface: a 14.5 km² box drawn at Ralegan Siddhi (Maharashtra), analysed from Copernicus GLO-30 — contours, streams, five ranked sites, the top site's 74 ha catchment and the results overlay (runoff 33 490 m³, storage 20 094 m³).

5.3 Database and Storage

Villages with PostGIS boundaries (an identical boundary reuses the village), DEM assets with provenance and statistics, jobs with idempotency keys, recommendations with a state machine whose transitions write a transactional outbox drained into an `audit_log` made append-only by database rules. Rasters (DEM, slope, TWI, flow accumulation, ...) are COGs in MinIO served as tiles by TiTiler; streams and siting are JSON documents next to them; the rainfall series is cached per point in the object store. On the lab VMs the same ports are backed by memory and local disk.

6 CSD THEMES AND TOPICS APPLIED IN THE PROJECT

Load balancing without shared state. The lab VMs cannot run PostgreSQL or Redis for this app, so replicas keep villages and jobs in memory; a user must keep reaching the replica that ran their analysis. `nginx ip_hash` gives that affinity, `max_fails` removes a dead replica and `proxy_next_upstream` retries on another — the failover a single VM cannot give (one lab VM's process died twice during the project). The cost is stated: a failed-over user re-runs a 2 s analysis. The Docker shape removes the limitation with shared PostGIS/Redis/MinIO. **Bulkheads over more threads.** The first lab load test showed `POST /analyzeArea` blocking 11–21 s: the inline runner ran the job inside the request. A thread-pool runner with separate *interactive* and *heavy* pools returns 202 in milliseconds; the heavy pool is deliberately small because the hydrology loops are GIL-bound Python — more threads would add waiting, not throughput;

Table 2. Mapping of core CSD themes/topics to their use in this project

CSD theme	Where used	Justification
API design (REST)	/api/v1, 202 + job id, problem details, generated OpenAPI	long work must not hold a connection; clients branch on stable codes
Load balancing	nginx ip_hash over four lab replicas, passive health checks	in-memory replicas need client affinity; a dead VM leaves rotation
Caching	single-flight rainfall cache; LRU GLO-30 windows; COG tiles; service worker	45-year series once per point; a cold herd costs one upstream call
Indexing / query optimisation	PostGIS geometry, unique idempotency key, COG internal tiling	village reuse by boundary; byte-range reads of only the needed window
Concurrency / async	bulkhead pools (interactive, heavy), WebSocket progress, 429 backpressure	a village analysis cannot delay a click; bounded queues under stress
Microservices vs. monolith	layered monolith + workers	one team, one deployable; boundaries enforced by a test, not a network
Design patterns	Ports & Adapters, Strategy, Decorator, Chain of Responsibility, Builder, Specification, Saga, State, Outbox, Observer	each solves a named problem (Section 3)
AuthN / AuthZ	RS256 JWT, three roles, officer-only approval	approvals are accountable; audit log is append-only
Containerisation / deployment	Docker Compose (11 services); one-process replicas on the lab VMs	same code, two shapes, chosen by configuration
Error handling / resilience	retry, circuit breaker, fallback chain, stale cache, stated defaults	degrade with a warning rather than fail
Algorithms and complexity	Priority-Flood $O(n \log n)$, topological sweep $O(n)$, reverse BFS, distance transform	exact, textbook, explainable; ~ 2 s per village
Testing strategy	219 tests: 41 golden (analytic terrain), API flows, pysheds cross-check, e2e over HTTP	hydrology correctness is checked against known answers
Version control / CI	conventional commits, GitHub Actions (ruff, mypy strict, pytest, image build)	every push proves the build

throughput comes from replicas. **Single-flight caching.** Ten users opening the same village all missed the rainfall cache and called Open-Meteo ten times (p95 43 s). A per-key lock makes concurrent misses wait for one upstream call (p95 63 ms).

7 RESULTS AND EVALUATION

Provided contour map (Khapri, Durg, Chhattisgarh; 1 355 contours, 267–298 m, SRTM 30 m source; UTM 44N derived from the file). The river entering from outside the map (the Shivnath) carries no in-map accumulation, so an early version placed the top site on its bank (Figure 7); with the water body mask, 92.7 ha of water and its 200 m belt are excluded and the top site is a tributary (Figure 3): catchment 103 ha $\pm 17\%$, 75 % dependable rainfall 1 192 mm at the site (the overlay’s 1 151 mm is the village-centroid series), CN 83, SCS-CN expected volume 172 126 m³ $\pm 30\%$ (rational 462 k, Strange 49 k: spread 182 %, reported), pond 50 000 m³ (size cap binds), 126 $\times$ 126 m $\times$ 3.5 m, fills in 25 of 25 years, INR 97 lakh indicative, confidence *moderate* (*low* when SoilGrids is unreachable and the soil group is assumed).

Validation. (i) 41 golden tests with analytic answers: inclined plane and cone interpolation (RMS < 0.25 m), pit filling to spill level, plane accumulation equal to the upslope cell count, V-valley catchment equal to the rectangle upstream, Y-valley Strahler order 2, edge-fed flags on a hill and a tilted plane, river and water exclusion with the flood belt, SCS-CN and frustum hand calculations. (ii) **Independent implementation:** pysheds [2] on the same DEM agrees within 2.0 % and 3.4 % on the main outlets (22.5 % on a floodplain flat where flat-resolution schemes differ). (iii) **Independent elevation source:** GLO-30 against the contour DEM on the sample — surfaces agree (RMSE 1.8 m after a -5 m datum offset, $r=0.94$) but on this flat terrain (mean slope 1.6°) small flow paths do not, and catchments at small sites differ several-fold (Figure 8); the algorithm is not the uncertainty, the source DEM is — hence the bands. (iv) Snapping cuts the pour-point coefficient of variation from 212 % to 48 %. (v) The two OSM-mapped tanks in the sample are canal-fed; local runoff is a tenth of their capacity — recorded honestly as a mismatch of purpose, not of method.

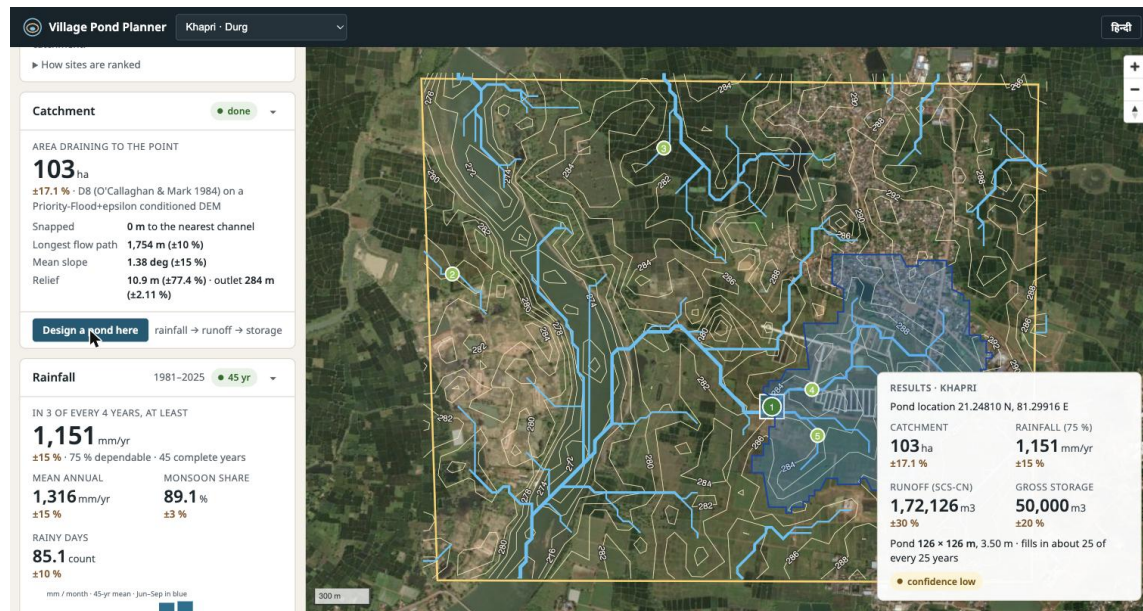


Fig. 3. Example results overlay for the provided contour map: ranked sites (all ≥ 295 m from mapped water), the top site's 103 ha catchment, and the pond result.

7.1 Performance

Measured with Locust, 20 concurrent users (10 clicking catchments, 10 drawing a *new* box each time, so no cache helps), 90 s, against one process (8-core laptop, thread-pool runner, live GLO-30 and Open-Meteo): POST /analyzeArea p95 17 ms; full area analysis (GLO-30 read, hydrology, siting, catchment) p50 2 s, p95 4 s, 154 analyses; catchment end to end p95 530 ms; rainfall p95 63 ms. Before the bulkhead runner and single-flight cache the same test gave 11–21 s and 43 s. The 46-check end-to-end HTTP suite passes against the deployed lab replica. On the Docker stack, 50 users clicking catchments gave POST p95 33 ms and end-to-end p95 560 ms with no failures; a catchment returned in 0.6 s while a heavy suitability job ran (bulkheads). **Limits within the four systems:** one process sustained 1.7 area analyses per second; the 25 km² cap bounds one request to seconds; 429 backpressure caps each queue at 20 jobs; replicas share nothing, so throughput should grow roughly linearly with the four lab replicas behind the balancer (expected, not yet measured on the lab network).

8 DISCUSSION AND LIMITATIONS

What worked: one hydrology chain behind a DEM port served both the Phase 2 upload and the Phase 3 map selection unchanged; every edge case found by looking at the map (a pond on the river, a truncated catchment, a flattened river surface) became a hard rule with a golden test and a named warning. **Limits:** both DEMs are 30 m (SRTM ± 6 m and GLO-30 ± 2 m relative); relief below a few metres is not real, and on flat land the catchment of a small site is dominated by DEM error — the reason for the ± 15 –50 % bands and the advice to verify the outlet in the field. GLO-30 is a surface model (trees and buildings included). Reanalysis rainfall is about ± 15 % against gauges. The three runoff methods disagree by up to 2 $\times$; SCS-CN is the design figure and the range is shown. Land ownership is unknown without

cadastral data and is never assumed. The ML scorer is deferred: two mapped tanks are not a training set. Lab replicas keep state in memory, so a restart forgets villages (re-analysis takes seconds).

9 AI TOOL USAGE DECLARATION

An AI coding assistant (Claude Code, Anthropic) was used under the author's direction and review, as the assignment's LLM policy permits, for: refactoring and structuring code into the layered design; brainstorming and understanding the algorithms (D8 as a functional graph, why descending elevation is a topological order, catchment as reverse BFS, the river and truncation edge cases); implementing features to the author's specification with tests alongside (including the map-selection adapter and the lab deployment); debugging defects found by running the system; and editing this report and its figures, reformatting it into this template. Every number in the report is produced by the code. Every design decision is recorded with its reasoning and rejected alternatives in 20 architecture decision records and the decision log (docs/PROGRESS.md); the author has reviewed, understood and can explain each component.

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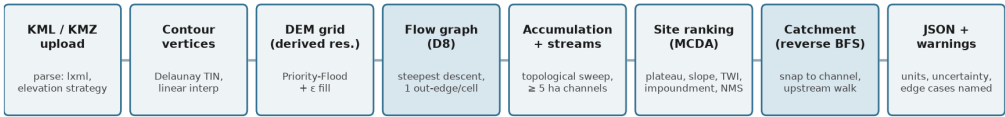
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A SOURCE CODE AND REPOSITORY

Repository: <https://github.com/Rahul5977/AI-BasedPondAnalysis> (public). **Working front-end:** <http://10.1.75.53:4270> (Swagger at /docs, planner at /app). **Demo video (5 min):** [YouTube link — add after upload]. **Run anywhere:** make
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up && make seed (Docker, 11 services) or make serve-single (one process); make check runs lint, types and tests.
Structure: app/ (api routers, schemas, engines — terrain, hydrology, rainfall, runoff, design, suitability, workflows —, domain, providers, repositories, jobs), web/ (React SPA), tests/ (incl. golden/), infra/ (Compose, Dockerfiles, nginx, lab replica script and load balancer), docs/ (ADRs, API cookbook and error catalogue, figures, progress log), scripts/ (figures, cross-checks, e2e suite).

B ADDITIONAL FIGURES



Highlighted stages are the graph algorithm: the D8 flow graph and the reverse breadth-first search over it.

Fig. 4. The analysis pipeline; the highlighted stages form the graph algorithm.

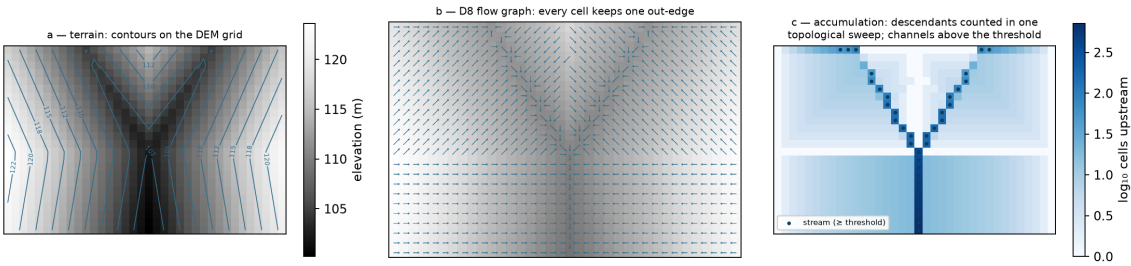


Fig. 5. DEM → D8 flow graph → accumulation and streams (generated from the engine code).

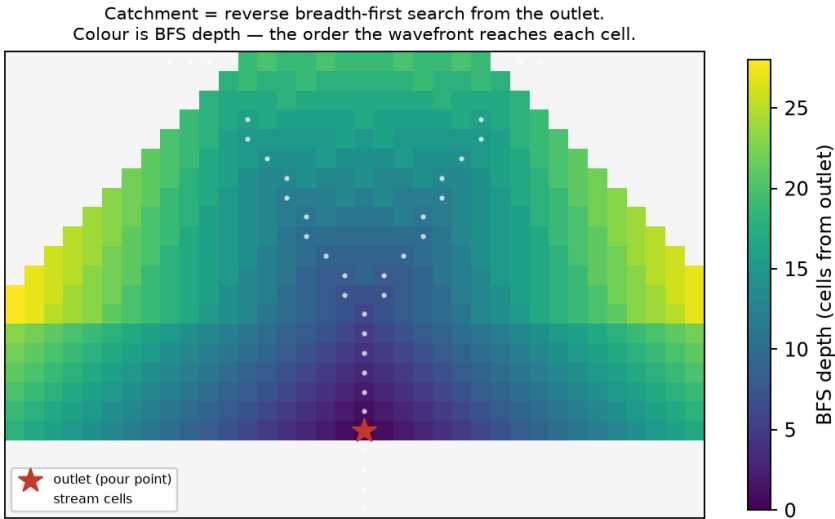


Fig. 6. Catchment as a reverse BFS from the outlet; colour is the wavefront order.

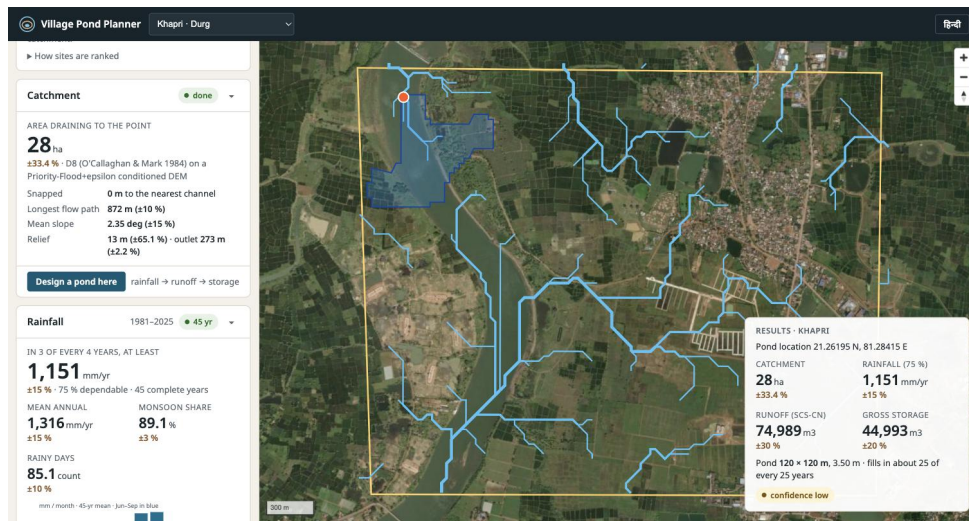


Fig. 7. Before the water body mask: the top site on the Shivnath's bank, its catchment 40 % permanent water. Compare Figure 3.

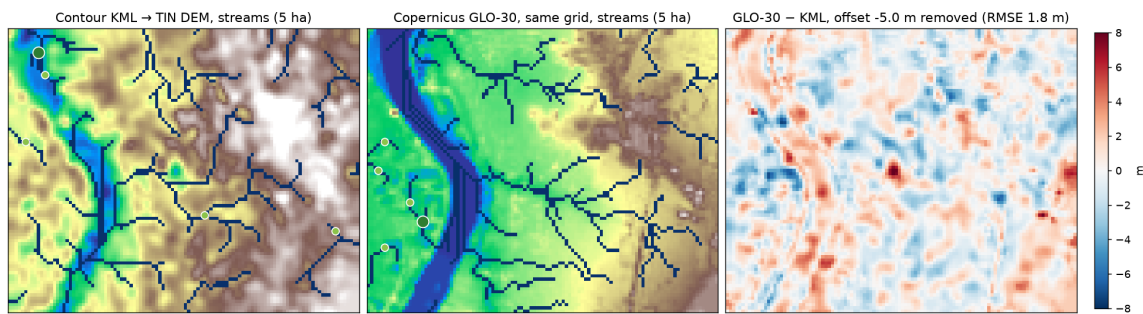


Fig. 8. Terrain-source cross-check on the sample: contour DEM vs. GLO-30 on the same grid, streams and sites from the identical chain, and the elevation difference.

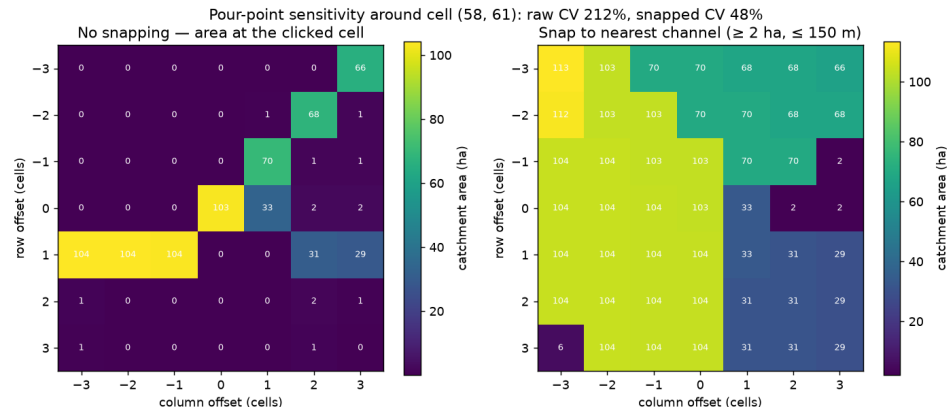


Fig. 9. Pour-point sensitivity: catchment area around a channel cell, raw vs. snapped.

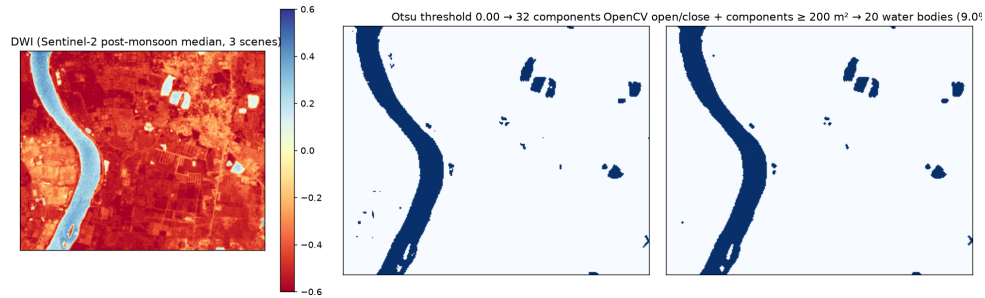


Fig. 10. NDWI water mask with OpenCV: Otsu threshold, morphology, connected components.